AMIANet: Asymmetric Multimodal Interactive Augmentation Network for Semantic Segmentation of Remote Sensing Imagery

Tiancheng Liu, Qingwu Hu, Wenlei Fan, Haixia Feng, and Daoyuan Zheng

***Abstract*— In recent years, the inherent 2-D characteristics of optical images have led to a plateau in semantic segmentation performance. The complementary nature of light detection and ranging (LiDAR) point clouds and camera images can effectively enhance semantic segmentation capabilities, and thus, research into multimodal joint semantic segmentation is garnering increasing attention. However, the domain gaps between different dimensions present challenges for the fusion of multimodal data. In this article, we introduce a novel asymmetric multimodal interaction augmented network (AMIANet), which directly processes heterogeneous data from images and point clouds. The treatment of the disparities in modal data ensures consistency in the features of both modes. Through the newly developed synergistic multimodal interaction module (SMI Module), AMIANet is capable of combining the complementary characteristics of cross-modal data. This is achieved by interactively fusing and extracting precise and rich structural information from point cloud features to enhance image characteristics. The experimental results on the N3C-California, WHU-RRDSD, and ISPRS Vaihingen datasets demonstrate that AMIANet surpasses benchmark methods and current state-of-the-art (SOTA) approaches. The code will be available at https://github.com/2012153946/AMIANet.**

***Index Terms*— Light detection and ranging (LiDAR), multimodal fusion, remote sensing imagery, semantic segmentation.**

I. Introduction

SEMANTIC segmentation technology is an increasingly critical tool in various fields, notably urban planning and smart agriculture [1]. The evolution of deep learning and advanced equipment has catalyzed the development of numerous efficient and high-precision methods in the domain of remote sensing image semantic segmentation. Predominantly, optical remote sensing images have become a primary data source, thanks to their rich semantic content and ease of acquisition [2]. Despite this, the limitations of single-modality imaging, such as challenges posed by poor lighting or severe occlusion, significantly hamper the accuracy of segmentation.

With the deepening research on multispectral and hyperspectral images in the field of remote sensing, the semantic segmentation of multisource information fusion has become increasingly popular. Many scholars have begun to explore semantic segmentation networks based on high-dimensional data [3], [4]. Although these high-dimensional data-based land cover classification methods can utilize rich spectral information to aid segmentation, the similarity in spectral characteristics among some land covers makes it difficult to distinguish them even with hyperspectral data, leading to bottlenecks in remote sensing image semantic segmentation [5].

Light detection and ranging (LiDAR) point cloud data, characterized by its abundant spatial structural information, has been pivotal in traditional segmentation methods [6], [7]. For instance, the distinct flatness characteristics of point clouds enable the rapid differentiation of vegetation, while the echo information assists in distinguishing between various forms of vegetation. In addition, the 3-D structural information accurately outlines building contours. However, the absence of texture information in point cloud data alone often leads to shortcomings in achieving intelligent semantic segmentation.

With in-depth research on multimodal data, combining image and LiDAR data for processing has become an important means to break the limitations of single modality. However, due to the significant differences in the organization of LiDAR point cloud and image data, many researchers have converted point clouds into 2-D products [such as normalized digital surface model (nDSM)] and then combined them with image data in various fusion classification models.

DFCN [8] combines the idea of dense connectivity with a full convolutional layer architecture and proposes a specific weighted fusion strategy to integrate optical images and digital surface model (DSM) data for multimodal remote sensing data segmentation. GMNet [9] upgrades multimodal features by grading and refining them, introducing special fusion modules for different feature levels to effectively utilize multimodal features. G2GNet [10] suggests an adaptive attention mechanism module to gather information features from multimodal data, merging various features into a set of global descriptors to enhance the RGB feature representation. CMGFNet [11] introduces a broad-modal gating fusion network, successfully segmenting buildings from very high-resolution (VHR) remote sensing images and DSM data. He et al. [12], inspired by neuropsychology, proposed a hypergraph parser to imitate guided perception for learning intramodal object relationships, uncovering semantic clues, and generating robust monomodal

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representations, and a hypergraph matcher to improve cross-modal compatibility in multimodal feature fusion. These methods choose to project laser point clouds into 2-D products for fusion with image data. Although they solve the problem of cross-modal data structural differences, they inevitably lose some of the most important spatial semantic information.

Recent advances in point cloud deep learning technology have led to methods that directly fuse raw point cloud data with images. Maiti et al. [13] first refined point cloud data features based on attention modules and then fused the extracted point cloud features with image features using projection matrices and attention fusion modules. PointBoost [14] proposed a sequential segmentation network that utilizes mature point cloud deep learning networks for feature extraction before fusing them with original images for input into image semantic segmentation networks. IKD-Net [15] introduced a parallel multimodal semantic segmentation network, which uses two types of gating modules to parallelize point cloud and image branches. Although these cross-modal methods avoid information loss caused by projection, they do not fully consider the imbalance in feature extraction processes across different modalities and fail to effectively utilize the complementary advantages of different modal data.

To bridge this gap, a novel framework, asymmetric multimodal interaction augmented network (AMIANet), is designed to accentuate the strengths of multimodal data in semantic segmentation. AMIANet performs initial feature refinement on LiDAR data, followed by a systematic bidirectional fusion of different modal features, enabling effective interaction between modes.

The main contributions of this article are as follows.

1) We present AMIANet, an efficient multimodal segmentation framework that leverages the synergistic multimodal interaction module (SMI Module) to facilitate bidirectional feature interaction. This framework is designed to ensure balanced data extraction from both modes, providing a comprehensive feature representation for image semantic segmentation tasks. The effectiveness of AMIANet has been validated on various multimodal datasets, highlighting its advantages in semantic segmentation.
2) The multiscale convolutional projection module (MCP Module) ensures data integrity during cross-modal data fusion and offers new insights into addressing data loss issues.
3) Based on the point-driven image fusion module (PDIF Module) and the image-driven point fusion module (IDPF Module), the bidirectional fusion of cross-modal features is achieved, ensuring the full utilization and balanced flow of information from both RS modes.

The remainder of this article is organized as follows. In Section II, we briefly describe the related work on semantic segmentation of point clouds and multimodal feature learning based on point clouds and images. In Section III, the design details and key components of the AMIANet are thoroughly explored. The experimental results are presented and analyzed in Section IV. Section V discusses the experimental results. Finally, Section VI provides a summary of this article.

II. Related Work

A. 3-D Semantic Segmentation Based on LiDAR Point Clouds

With the rapid development of deep learning technology, deep neural networks have become a dominant technique for various image processing tasks. However, due to the significantly different data organization method of point clouds compared to images, many deep learning frameworks targeting the 3-D structure of point clouds have been successively proposed [16], [17], [18].

Previous methods primarily rely on projection transformation. This technique involves projecting point clouds into 2-D images from various angles. Such a process enabled the application of established deep learning models for image classification. Subsequently, the classification outcomes were reprojected onto the point clouds, leading to the acquisition of 3-D segmentation results [19], [20], [21]. Voxel-based approaches typically entail organizing point cloud data via a 3-D grid system. The data are then processed through 3-D convolution to perform feature extraction and classification on these voxels [22], [23], [24], [25]. However, these methodologies fall short in effectively leveraging the spatial structural information inherent in point clouds, and they face challenges in simultaneously recognizing and classifying geographical features across diverse scales.

Point-based methods effectively utilize the semantic information of point clouds for feature extraction. Charles et al. [26] pioneered PointNet, which employs multilayer perceptrons to map raw point clouds to a high-dimensional feature space. It uses symmetric functions to address the unordered nature of point clouds, enabling simple and effective learning of 3-D representations of objects. However, it lacks extraction of regional semantic features. Building on this, Qi et al. [27] introduced PointNet++, which employs a hierarchical network structure to overcome this limitation. Jiang et al. [28] proposed PointSIFT, inspired by the SIFT operator in image processing. It captures regional semantic features from eight directions, enhancing the capability to capture 3-D shapes by stacking features from multiple directions. In addition, Yousefhussien et al. [29] developed a fully convolutional network structure based on PointNet to accomplish the task of extracting airborne point clouds.

RandLA-Net, proposed by Hu et al. [30], demonstrates significant advantages in processing large-scale point clouds. Owing to its random sampling strategy and local feature aggregation (LFA) module, it maintains complete spatial semantic features while enhancing network efficiency, leading to many successful studies based on RandLA-Net [31], [32], [33], [34], [35], [36].

B. 2-D Semantic Segmentation Based on Multimodal Data

Point clouds can compensate for the lack of spatial structural semantic information in 2-D images. In recent years, several network architectures using multimodal data as input have emerged [37], [38], [39], [40]. Due to the structural and organizational differences between point cloud data and images, most deep learning methods for multimodal data adopt the approach of converting 3-D laser point clouds into 2-D products.

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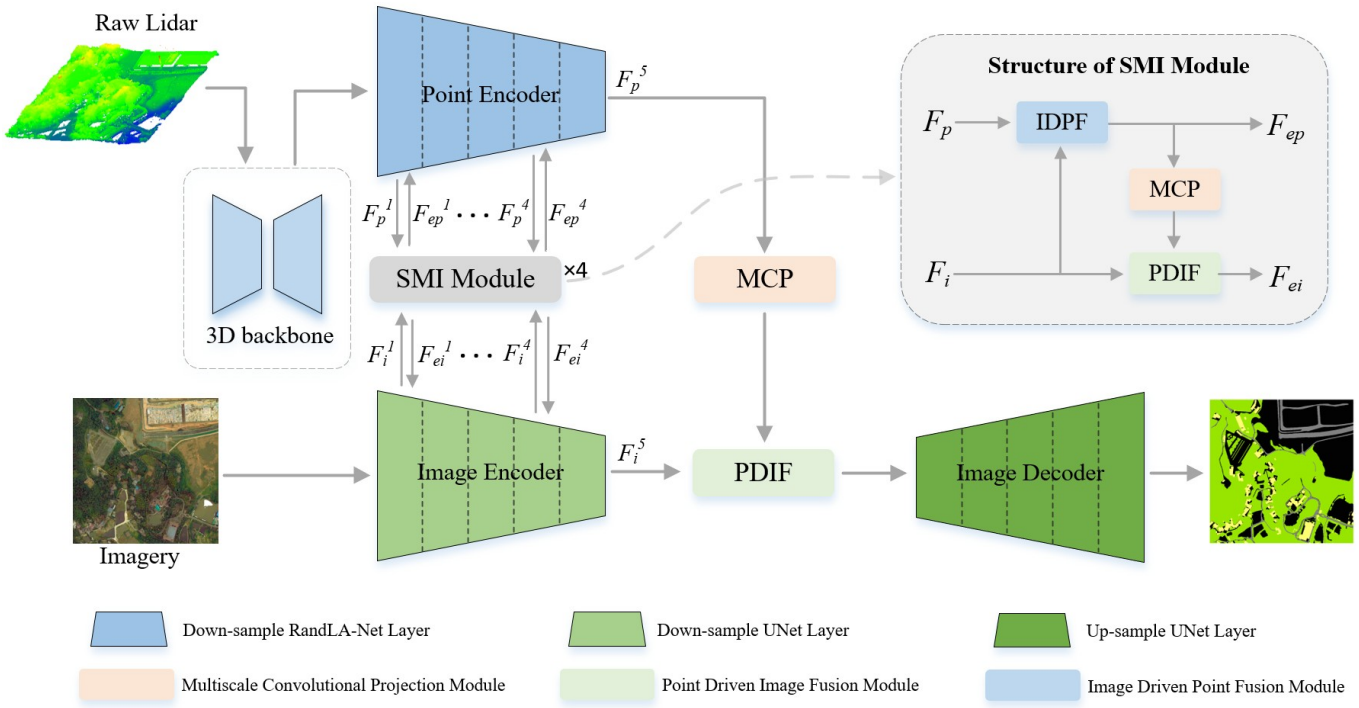

Fig. 1. Workflow of AMIANet.

Eitel et al. [41] utilized two separate convolutional networks to process RGB and depth information independently, merging them later to demonstrate the advantages of multimodal networks. Hazirbas et al. [42] introduced FuseNet, a parallel SegNet structure network, which independently extracts features in parallel in the encoder part and integrates depth information into RGB features in the decoder part. Audebert et al. [43] pointed out the disadvantages of applying FuseNet to remote sensing data and proposed an efficient multiscale convolutional neural network combining SegNet and ResNet. Nahhas et al. [44] filtered point clouds to generate DSM, DEM, nDSM, and echo count information, which, along with RGB information, was fed into a convolutional network for feature fusion and abstraction. Sun et al. [45] implemented adaptive fusion of multimodal data using attention mechanisms and pyramid dilation blocks and designed a multiscale supervision training scheme to improve the network performance.

Hosseinpour et al. [11] proposed a gated fusion module to address the issue of cross-modal feature fusion, combining it with depthwise separable convolutions to reduce network parameters. Fan et al. [46] introduced PACSCNet, which leverages the information advantage of multimodal data by selectively utilizing complementary information from different modes, incorporating an inverted pyramid residual structure to refine features from various modalities. These multimodal methods hold significant advantages over unimodal networks but inevitably suffer from spatial information loss due to the projection step.

Recent studies have attempted to directly combine point cloud deep networks with image network feature extraction processes. Maiti et al. [13] initially extracted features separately from point clouds and images, and then fused the modal features via projection matrices to ensure the integrity of point cloud spatial information. Zhang et al. [14], in their PointBoost proposal, first input point cloud data into a feature extraction network to obtain deep semantic information, which is then combined with RGB data for feature extraction and land object classification in a UNet network. Wang et al. [15] utilized gating and projection modules for cross-modal information fusion, employing point clouds to refine weaker image features in global and classification tasks. These approaches have reduced spatial semantic information loss in point cloud cross-modal fusion but have not fully utilized the complementary information of multimodal data. In addition, due to the organizational characteristics of point cloud data, symmetric parallel networks may lead to imbalanced information fusion.

III. Method

The application of multimodal data in the interpretation of geographical features is an established method to enhance accuracy, especially under challenging conditions such as poor image quality or extensive coverage and occlusion. Common practice involves transforming LiDAR data into a 2-D format for integration with imagery in 2-D networks, albeit at the cost of significant spatial structural information. Importantly, spatial structure, a fundamental attribute of point clouds, was not efficiently extracted by early point cloud networks. Consequently, running these networks in parallel with image networks creates an imbalance in the feature information extracted at each stage, thus underutilizing the rich information resources available in multimodal data. To overcome these limitations, AMIANet is introduced, as depicted in Fig. 1. The framework commenced with a spatial feature extraction from LiDAR data before the parallel symmetrical backbones for LiDAR and imagery, ensuring a simultaneous acquisition of rich feature information during

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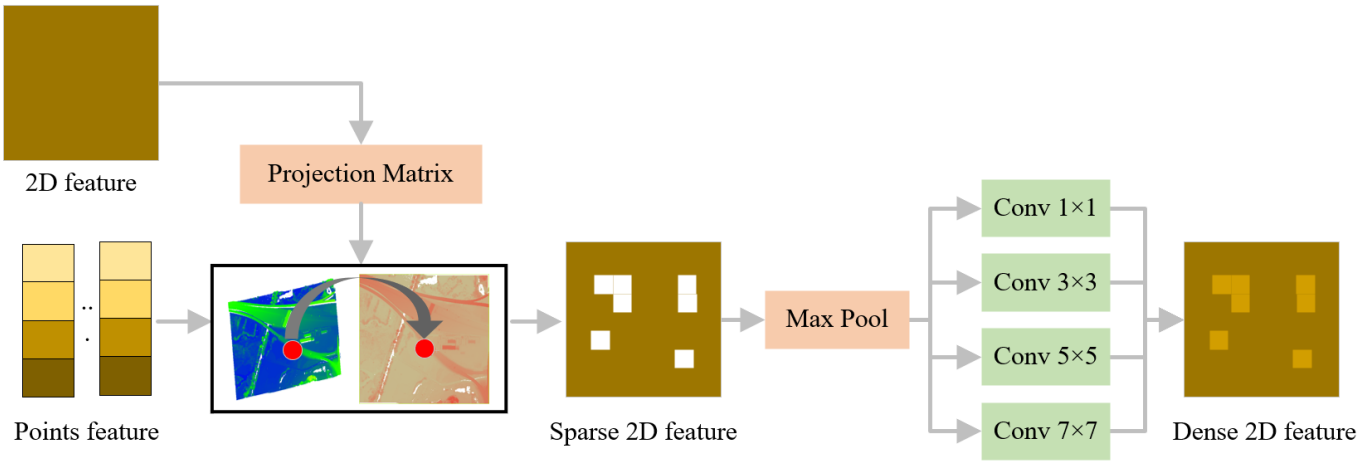

Fig. 2. Structure of the multiscale convolutional projection module.

the encoder phase. Building on this, two novel feature fusion modules, the image-driven point feature (IDPF) module and the point-driven image feature (PDIF) module, are proposed to facilitate bidirectional feature enhancement. These modules collectively form the SMI module, which initially employs the IDPF module to bolster the semantic information of point clouds using image features, thereby generating enriched point cloud features. Subsequently, the PDIF module integrates these enhanced features, producing image features with enriched spatial information. The multiscale convolutional projection (MCP) module is introduced to support the matching process in the PDIF module, with reverse matching achieved through coordinate position projection.

A. Asymmetric Network Structure

The asymmetric network architecture primarily consists of a LiDAR feature extraction backbone, two parallel symmetrical encoders for LiDAR and imagery, and an imagery stream decoder. Compared to imagery, LiDAR data possess more complex feature information, and due to the unordered nature of point clouds, the feature extraction process for point cloud data is more intricate. In a symmetric structure, it is challenging to balance these features with image features, particularly when the initial features in point cloud data are sparse, as this imbalance can negatively impact the refinement of image features. To address this issue, spatial features are initially extracted from the LiDAR point clouds before being fed into the point cloud encoder. The method for preliminary feature extraction from point clouds is not strictly defined and can be selected based on the specific characteristics of the point clouds and application scenarios. In this study, RandLA-Net is used for point cloud feature extraction. The RandLA-Net structure follows an encoder–decoder paradigm with skip connections. The encoding part utilizes an LFA module to learn spatial features of the point cloud and employs a random sampling algorithm for point cloud downsampling, discarding the final three fully connected (FC) layers and the dropout layer to preserve feature information. In the point encoder, multiple SMI modules facilitate multimodal feature communication, and the PDIF module is used at the end of the encoder to ensure efficient feature fusion. To align the features' output at the same depth in the point cloud branch of the encoder with the image features, the encoder for point cloud feature extraction still employs RandLA-Net based on random sampling. The encoder for the image branch and the final decoder are derived from UNet [47]. It is important to note that in the decoder part, only the essential image stream is maintained, discarding the point cloud branch, as the interpolation method used in the decoder to restore resolution affects feature alignment. Furthermore, since the feature fusion in the encoder part already fully exploits the advantages of multimodal data, removing the point cloud branch in the decoder part allows for significant efficiency gains with minimal sacrifice in precision.

B. Multiscale Convolutional Projection Module

To address the registration issue of multimodal data, an MCP module is developed. As shown in Fig. 2, a projection transformation is initially performed on the LiDAR data, thereby generating 2-D features that are of the same dimension as the imagery features at the corresponding layer level. Due to the discrete nature of point cloud data, features generated via projection transformation inevitably contain missing values. To mitigate this, an MSP approach is employed to expand the feature map.

The MCP module consists of a max-pooling layer and a multiscale feature extraction block. The latter is composed of convolutional kernels of various sizes working in parallel. The max-pooling layer preliminarily fills in the gaps in the feature map, similar to the dilation operation in image processing. A larger receptive field captures more pixel information, ensuring the integrity of feature learning. This parallel structure captures objects of different sizes for better feature representation. After processing by this module, point cloud features are projected into the image space for fusion. Conversely, image features are transferred to point cloud space without missing values, thus requiring only a coordinate-based transformation.

C. Point-Driven Image Fusion Module

The features of point clouds projected into the image space offer robust spatial support for the image stream. While most

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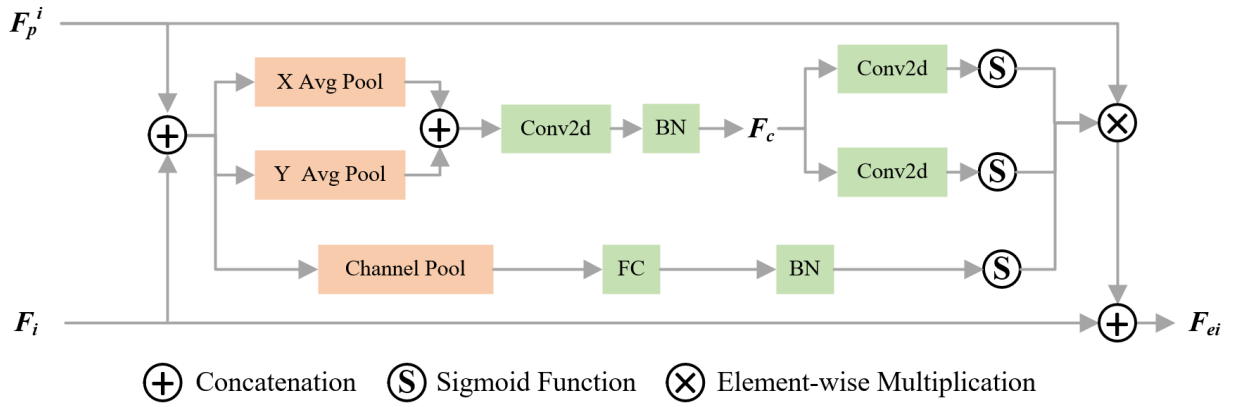

Fig. 3. Structure of the PDIF module.

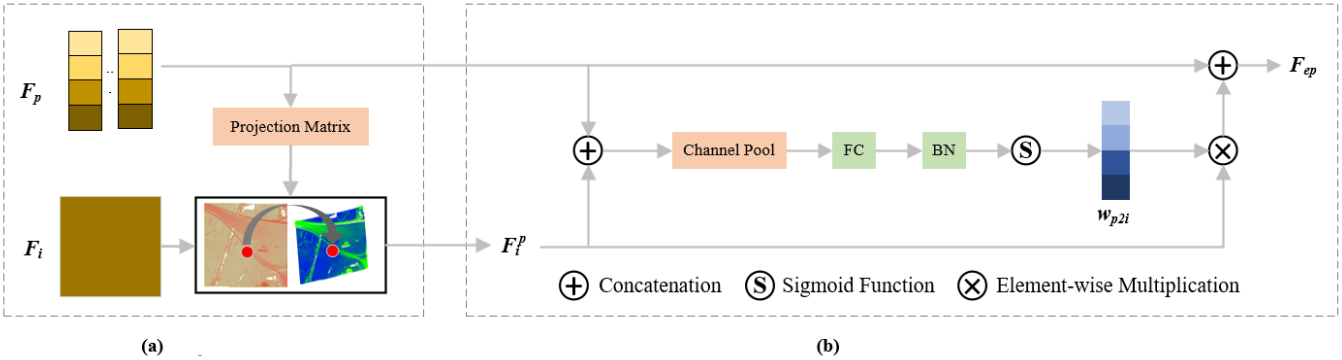

Fig. 4. Structure of (a) projection module and (b) IDPF module.

methods adopt a simple and effective direct concatenation approach for fusion, this can introduce noise and low-level features, especially when the original point cloud lacks color or texture attributes. To address this, the PDIF module is proposed to enhance the positional information in point cloud features and capture long-range dependencies within features. It utilizes spatial weights derived from multimodal data to suppress detrimental information in point cloud features and enhance beneficial aspects. The structure of the PDIF module is shown in Fig. 3. The projected point cloud feature F_p^i and the image feature F_i are concatenated to form the fused feature F_m, which is then input into coordinate and channel attention modules. The coordinate attention module compresses the feature map along the x-axis and y-axis to obtain spatial position encoding weights. This involves averaging and concatenation along the respective axes, followed by a shared 1×1 convolution, the output of which is denoted as F_c

$$F_c = \mu\left(\text{Conv}_{1\times1}\left(\left[X\text{AvgPool}(F_m) Y\text{AvgPool}(F_m)\right]\right)\right) \quad (1)$$

where μ represents the nonlinear activation function; [,] denotes the concatenation operation; and XAvgPool and YAvgPool refer to average pooling operations along the horizontal and vertical axes, respectively. The calculation process for the resulting spatial encoding weight $W_{\text{coordinate}}$ is as follows:

$$W_{\text{coordinate}} = \sigma\left(\text{Conv}_{1\times1}\left(F_c^h\right)\right) \otimes \sigma\left(\text{Conv}_{1\times1}\left(F_c^w\right)\right) \quad (2)$$

where σ represents the sigmoid activation function and F_c^h and F_c^w are the horizontal and vertical tensors, respectively. Within the channel attention module, a scheme akin is employed in the convolutional block attention module (CBAM) [48]. The channel attention weight W_{channel} is defined as follows:

$$W_{\text{channel}}^i = \sigma\left(\text{Conv}_{1\times1}\left(\left[\text{AvgPool}(F_m), \text{MaxPool}(F_m)\right]\right)\right). \quad (3)$$

Based on the weights derived from the two types of multimodal data, the projected point cloud features are enhanced. Subsequently, the enhanced point cloud features are concatenated with the image features. Therefore, the output of the PDIF module can be described as follows:

$$F_{ei} = \left[F_p^i \otimes W_{\text{channel}}^i \otimes W_{\text{coordinate}}, F_i\right]. \quad (4)$$

The PDIF module, through the use of both coordinate and channel attention modules, enhances the spatial features of point clouds while enriching feature details using image features, thereby fully leveraging the advantages of multimodal data.

D. Image-Driven Point Fusion Module

The aforementioned PDIF module utilizes point cloud features to enhance the semantic information of images. While effective, the PDIF module facilitates only a unidirectional fusion path from point cloud to image, which may not fully exploit the complementarity between these two modalities. We posit that the reverse path, from image to point cloud,

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is also valuable. Since point clouds inherently lack features such as texture, combining them with image features can help distinguish objects with similar spatial information. Based on these characteristics, the IDPF module is introduced to enable the feature enhancement path from image to point cloud. The aim of the IDPF module is to enhance the robustness of the fusion process. Given that image information can be compromised by occlusions and cloud interference, direct fusion could adversely affect the accuracy of point cloud features. Therefore, an attention mechanism is designed to calculate the correlation between image and point cloud features to avoid inaccuracies from image features. As shown in Fig. 4(a), we first complete the conversion from the image dimension to the point cloud dimension based on a projection module similar to the MCP module. Subsequently, as illustrated in Fig. 4(b), similar to the channel attention module in the PDIF module, the attention weight W^p_{channel} in the IDPF module can be expressed as follows:

$$W^p_{\text{channel}} = \sigma\left(\text{Conv}_{1\times1}\left(\left[\text{AvgPool}(F_m), \text{MaxPool}(F_m)\right]\right)\right) \quad (5)$$

where $F_m = [F_p, F_i^p]$.

Therefore, the output of the IDPF module can be represented as follows:

$$F_{\text{ep}} = \left[F_i^p \otimes W^p_{\text{channel}}, F_p\right]. \quad (6)$$

As illustrated in Fig. 4(b), the alternating combination of two modules forms a bidirectional multimodal fusion module. This module first utilizes the IDPF module to enhance the semantic information of the point cloud using image features, generating richer point cloud features. It then employs the PDIF module to merge the enhanced point cloud features, outputting image features enriched with comprehensive spatial information.

IV. Experiments

In this section, the experimental setup is detailed. First, three different types of datasets used in the experiments are introduced. Then, the hardware specifications and parameter settings utilized in the experiments are described. Finally, the results of the experiments on different datasets are presented, demonstrating the quantitative and qualitative performance of the proposed method in comparison to benchmarks and other state-of-the-art (SOTA) methods.

A. Dataset Description

1) N3C-California: The N3C-California dataset [15] comprises 1212 pairs of LiDAR and image tiles across four semantic categories (ground, trees, buildings, and others), covering most residential areas in Santa Clara County, California. For deep learning tasks, this dataset is divided into 10 800 image blocks of 512 × 512 pixels, with a resolution of 100 cm/pixel. The dataset is randomly split into training, validation, and test sets in an 8:1:1 ratio.

2) WHU-RRDSD: The WHU-RRDSD (sample database of control elements of railway route design) dataset [60] encompasses areas around the railway section from Changsha to Ganzhou City, passing through Pingxiang, covering approximately 70 km^2. The research area includes typical Chinese rural buildings, diverse vegetation types, and various scales of highways.

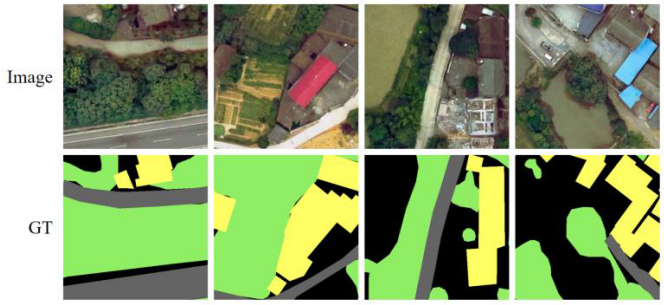

Fig. 5. Examples of WHU-RRDSD datasets.

WHU-RRDSD dataset contains DOM, digital line graphs, and LiDAR point clouds, offering robust data support for a variety of deep learning tasks. In the context of multiclass semantic segmentation, the dataset provides labels for four semantic categories: buildings, vegetation, roads, and others. The dataset comprises 7447 image blocks, each with dimensions of 512 × 512 pixels and a spatial resolution of 100 cm/pixel. Fig. 5 illustrates several representative examples from the dataset. However, due to the inherent limitations of the data collection equipment, some regions within the LiDAR point cloud data exhibit noise and low density. As a result of the incompleteness of point cloud data in certain areas, full coverage of all 2-D data cannot be achieved. Consequently, 4000 samples that meet the requirements of the multimodal experiments are selected. These samples were subsequently partitioned into training, validation, and test sets in an 8:1:1 ratio.

3) ISPRS Vaihingen: The ISPRS Vaihingen dataset is an open dataset released by the official ISPRS, covering areas with dense buildings. The dataset includes IRRG synthesized false-color remote sensing images and DSM data, with an average size of 2493 × 2063 pixels and a resolution of 9 cm/pixel, featuring six types of land cover labels: buildings, trees, low vegetation, background, cars, and impervious surfaces. In addition, we adopted the same dataset division method as in [14].

B. Experimental Setup and Evaluation Metrics

All experiments in this study were conducted on an NVIDIA GeForce RTX A6000 48 GB GPU using the PyTorch framework. During the point cloud extraction phase, 131 072 points from each point cloud patch are randomly selected as input samples for the point cloud feature preprocessing module. This sampling strategy was designed to balance computational efficiency with the retention of critical information necessary for effective learning and inference. The chosen point count corresponds to half the total number of pixels in a 512 × 512 input image, ensuring that the sampled point sets exhibit comparable resolution and spatial distribution, thereby facilitating the model's processing tasks. During the training phase, we adhered to a strategy analogous to [15]. A consistent parameter setting was used across all three datasets, with a total of 50 000 training iterations and a batch size of 2.

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TABLE I

QUANTITATIVE COMPARISON OF AMIANET, THE BASELINE METHODS, THE MULTIMODAL BENCHMARK METHOD, THE VISUAL CLASSICAL SEMANTIC SEGMENTATION METHODS, AND THE SOTA MULTIMODAL SEGMENTATION NETWORKS ON THE N3C-CALIFORNIA DATASET

Method	Input	OA	mAcc	IoU				
				Others	Ground	Tree	Building	Mean
UNet [47]	RGB	86.35	67.17	2.46	80.92	73.33	81.01	59.43
RandLA-Net [30]	LiDAR	87.49	85.78	40.06	88.32	84.19	69.98	70.64
Hybri-UNet [49]	RGB+DSM	89.00	71.76	12.98	85.86	77.03	87.13	65.75
vFuseNet [43]	RGB+DSM	86.11	75.47	**49.73**	81.99	57.58	77.43	66.68
SGFNet [50]	RGB+DSM	91.75	80.31	36.92	73.17	80.46	82.62	68.29
MultifilterCNN [51]	RGB+DSM+intensity +number returns + DoG	88.97	76.44	25.77	84.33	77.63	82.63	67.59
MFNet [45]	RGB+DSM+Slop angle + DoG	91.00	74.85	14.29	87.36	82.09	89.87	68.40
MDL_RS [52]	RGB+DSM+intensity +number returns	90.99	72.55	16.46	87.08	74.82	86.57	66.23
JSH-Net [53]	RGB+DSM+intensity +number returns	91.59	75.46	23.56	88.00	75.18	88.23	68.74
EndNet [54]	RGB+DSM+intensity +number returns	88.29	66.58	3.11	8.34	72.23	78.87	59.40
CMGFNet [11]	RGB+DSM+intensity +number returns	92.90	75.95	22.69	93.65	80.53	92.90	72.44
PointBoost [14]	RGB + LiDAR	92.52	82.20	34.67	89.44	**87.65**	91.87	75.91
IKD-Net [15]	RGB + LiDAR	93.81	90.61	30.68	93.11	82.35	95.87	75.50
AMIANet (ours)	RGB + LiDAR	**95.27**	**92.84**	37.29	**94.23**	84.95	**96.04**	**78.13**

Therefore, the maximum numbers of training epochs on the WHU-RRDSD dataset, the N3C-California dataset, and the ISPRS Vaihingen dataset are 31, 12, and 120, respectively. The model was optimized using the Adam optimizer with a fixed learning rate of 0.001.

The results were evaluated using overall accuracy (OA), accuracy (Acc), and intersection over union (IoU). The computation methods for OA, Acc, and IoU are as follows:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{FN} + \mathrm{FP} + \mathrm{TN}} \tag{7}$$

$$\mathrm{Acc} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{8}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{9}$$

where TP, TN, FN, and FP, respectively, represent true positives, true negatives, false negatives, and false positives.

C. N3C-California Dataset Results

On the N3C-California dataset, we compared our AMIANet with baseline methods such as UNet and RandLA-Net [30], multimodal benchmark method Hybri-UNet [49], as well as various SOTA multimodal segmentation networks in remote sensing, including vFuseNet [43], SGFNet [50], MultifilterCNN [51], MFNet [45], MDL_RS [52], JSH-Net [53], EndNet [54], CMGFNet [11], PointBoost [14], and IKD-Net [15]. The results are presented in Table I, showcasing the performance comparisons across these various methods.

As evident from Table I, the proposed method demonstrates a significant advantage in identifying the "Tree" and "Building" categories. This advantage is likely attributable to the distinct and crucial spatial features of these categories, which are effectively utilized by the proposed approach. The "Others" category shows a lower IoU due to its nonuniform feature description; however, our method still performs commendably. For the "Ground" category, which can be adequately extracted based on planar features, our method further improves distinction from other categories by incorporating spatial information, thus achieving the highest accuracy. Overall, the proposed method outperforms other algorithms in detailed metrics, demonstrating a clear advantage.

The baseline method, RandLA-Net, scores higher than other classic multimodal segmentation methods in mAcc and IoU, possibly due to spatial information loss during the preprocessing stage in these methods. AMIANet shows superior performance compared to the SOTA multimodal segmentation methods across all metrics. Compared to methods based on 2-D products, our approach shows an improvement of 16.4% in mAcc and 5.7% in mIoU. Even against the latest point cloud-based multimodal methods, AMIANet exhibits the best performance across all metrics. The superior results of AMIANet are mainly attributable to its asymmetric network structure and SMI module. The asymmetric structure ensures a balanced information exchange, while the interaction module provides a complementary fusion platform for multimodal features.

Fig. 6 visualizes the qualitative comparison results of AMIANet and several other methods in six different scenarios.

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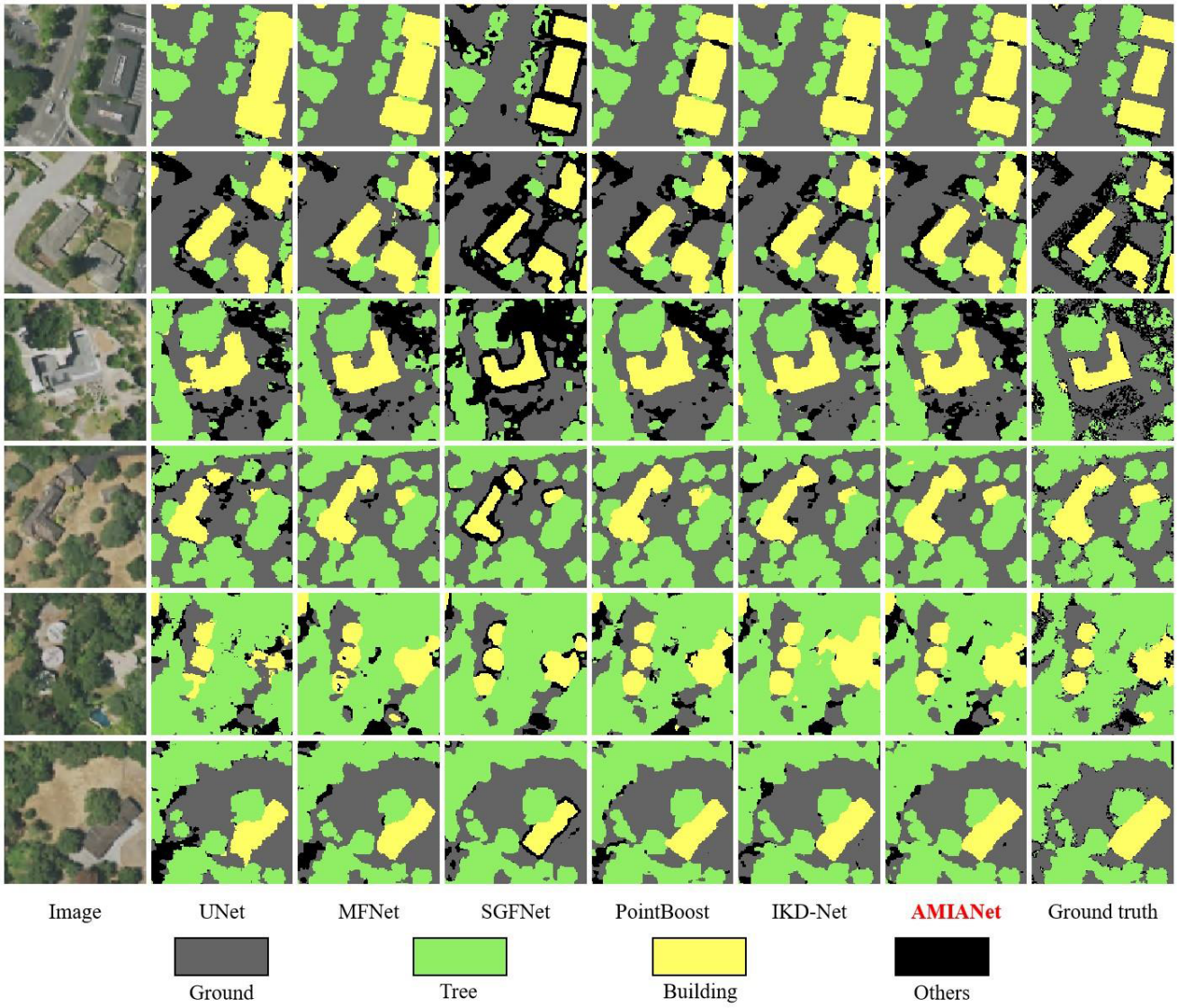

Fig. 6. Qualitative comparison of the baseline, the SOTA multimodal segmentation networks, and AMIANet (ours) on N3C-California.

UNet often fails to perform segmentation tasks effectively in many scenarios. For instance, in the fifth and sixth scenes, methods based on a single modality struggle to correctly differentiate categories in the presence of occlusions or shadows. On the other hand, multimodal segmentation methods based directly on LiDAR are relatively more adept at delineating the edges of buildings, with AMIANet also ensuring accurate extraction of vegetation areas.

To better understand the impact of each module on feature refinement, the feature maps after each module's operation in the first layer of the structure are visualized in Fig. 7.

From each row in the figure, it can be seen that after processing through each module, the features are progressively refined. Specifically, after the MCP module's operation, the feature maps exhibit fewer gaps and discrete elements in the central regions. However, at this stage, the features primarily capture object contours and edges, which limits their ability to effectively distinguish between buildings and vegetation. In contrast, following the feature fusion in the PDIF module, more comprehensive information was utilized, enabling a clearer distinction between buildings and vegetation.

D. WHU-RRDSD Dataset Results

In the WHU-RRDSD dataset, the point cloud data are relatively sparse, and the spatial structures of the objects to be classified are similar, allowing for a comparison of the classification capabilities of different methods under challenging data conditions. On the WHU-RRDSD dataset, AMIANet was compared with baseline methods such as UNet, HRNet [55], and RandLA-Net, the multimodal benchmark method Hybri-UNet, and SOTA multimodal segmentation networks, including MFNet, ESANet [56], DRNet [57], SGFNet, PointBoost, and IKD-Net. The results are displayed in Table II, illustrating the performance of these methods in this particular dataset context.

The results indicate that most methods exhibit poor performance in identifying the "Road" category, with the baseline method RandLA-Net showing a significant decline in

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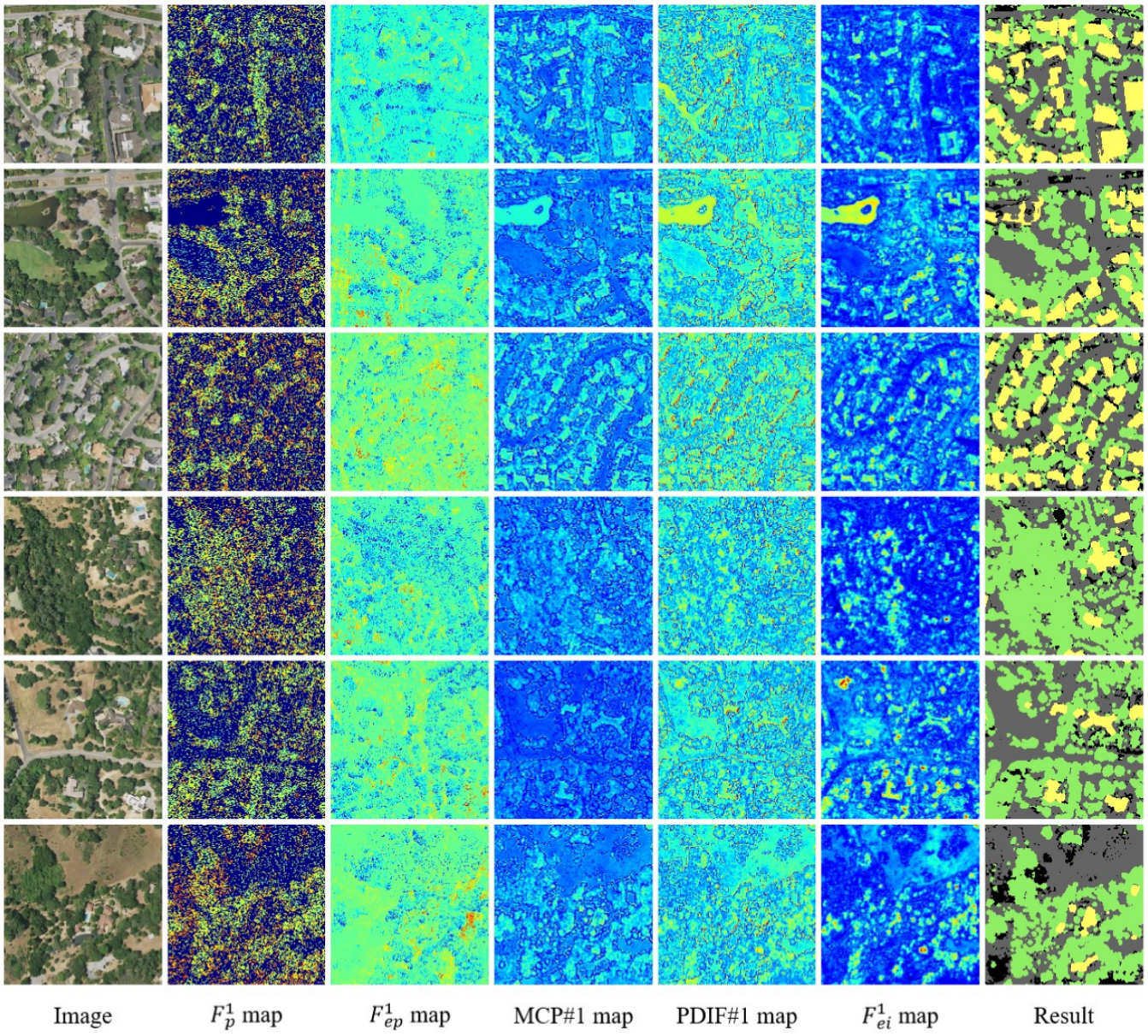

Fig. 7. Visualization of the feature maps from the first convolution layer of each module.

classification accuracy for this category. Similarly, the SOTA multimodal segmentation method based on LiDAR, IKD-Net, also shows lower precision than classic multimodal methods in this category. This is attributed to the small proportion of this category and its simple spatial structure, which easily confuses it with the "Others" category, where simple feature fusion does not significantly enhance the performance. In contrast, AMIANet, with its complementary feature fusion mechanism, achieves a notably higher IoU for the "Road" category.

AMIANet performs best in several comprehensive metrics. The advantage lies in its unique spatial structure, allowing for perfect complementarity of information from two different modalities during feature fusion. This avoids the impact of lower quality LiDAR data and enhances the model's robustness.

Fig. 8 showcases the segmentation results of AMIANet and several SOTA multimodal segmentation methods in five scenes from the WHU-RRDSD dataset. Multimodal methods, compared to the single-modal baseline method UNet, demonstrate superior classification capabilities, extracting the boundaries of buildings and vegetation more completely. In the second scene, classic multimodal segmentation methods encounter difficulties in accurately classifying buildings, whereas LiDAR-based multimodal methods excel in extracting precise target boundaries. Among these, AMIANet not only achieves more accurate road separation but also avoids the common misclassifications seen in other methods.

E. ISPRS Vaihingen Dataset Results

On the ISPRS Vaihingen dataset, AMIANet was compared with several baseline methods such as UNet, SegNet [58], and HRNet, as well as advanced multimodal segmentation networks, including SFAFMANet [59], ESANet, DRNet,

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TABLE II

QUANTITATIVE COMPARISON OF AMIANET, THE BASELINE METHODS, THE MULTIMODAL BENCHMARK METHOD, THE VISUAL CLASSICAL SEMANTIC SEGMENTATION METHODS, AND THE SOTA MULTIMODAL SEGMENTATION NETWORKS ON WHU-RRDSD DATASET

Method	Input	OA	mAcc	IoU				
				Others	Building	Road	Vegetation	Mean
UNet [47]	RGB	78.79	61.37	51.6	65.25	32.08	76.01	56.24
RandLA-Net [30]	LiDAR	70.69	65.97	40.87	72.23	15.11	74.22	50.61
Hybri-UNet [49]	RGB+DSM	81.38	65.12	52.1	67.11	35.09	76.09	57.60
HRNet [55]	RGB	83.16	63.79	55.32	68.11	36.17	77.57	59.29
ESANet [56]	RGB+DSM	84.47	70.25	57.19	61.36	46.84	82.93	62.08
DRNet [57]	RGB+DSM	83.35	69.14	55.69	65.76	39.14	79.31	59.97
MFNet [45]	RGB+DSM+ Slop angle + DoG	84.76	68.63	54.25	70.35	34.97	77.14	59.18
SGFNet [50]	RGB+DSM	87.1	72.82	61.89	63.51	**50.97**	85.91	65.57
PointBoost [14]	RGB+LiDAR	86.14	71.75	**62.71**	74.96	36.34	82.36	64.09
IKD-Net [15]	RGB+LiDAR	82.07	70.64	57.66	75.24	29.4	85.84	62.04
AMIANet (ours)	RGB+LiDAR	**87.44**	**74.84**	60.49	**77.45**	44.35	**87.73**	**67.51**

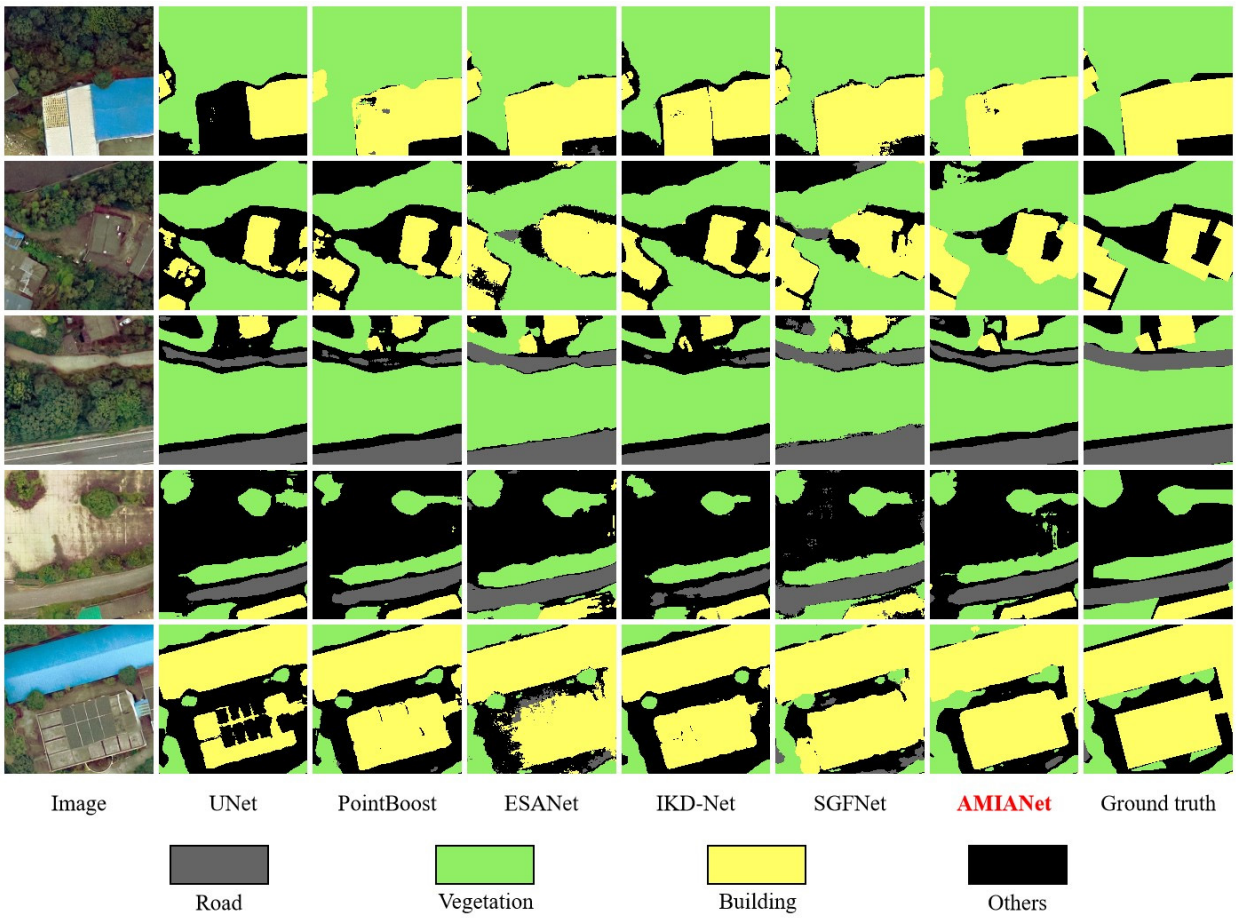

Fig. 8. Qualitative comparison of the baseline, the SOTA multimodal segmentation networks, and AMIANet (ours) on WHU-RRDSD.

PACSCNet [46], SGFNet, PointBoost, and IKD-Net. The comparison results are presented in Table III.

As shown in Table III, our method demonstrated significant advantages in recognizing the "Imp surf" and "Building"

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TABLE III
QUANTITATIVE COMPARISON OF AMIANET, THE BASELINE METHODS, THE MULTIMODAL BENCHMARK METHOD, THE VISUAL CLASSICAL SEMANTIC SEGMENTATION METHODS, AND THE SOTA MULTIMODAL SEGMENTATION NETWORKS ON THE ISPRS VAIHINGEN DATASET

Method	Input	OA	IoU					
			Imp surf	Building	Low veg	Tree	Car	Mean
UNet [47]	RGB	84.75	72.01	80.52	56.47	70.42	46.27	65.14
SegNet [58]	RGB	86.02	71.54	79.27	54.41	69.93	41.72	63.37
HRNet [55]	RGB	86.78	74.45	81.66	56.42	70.43	46.74	65.94
SFAFMANet [59]	RGB+DSM	87.19	77.36	88.05	61.25	73.77	57.34	71.55
ESANet [56]	RGB+DSM	89.79	78.32	87.58	61.73	74.02	62.18	72.77
DRNet [57]	RGB+DSM	90.89	80.34	88.42	62.31	74.25	67.18	74.5
PACSCNet [46]	RGB+DSM	90.11	79.93	88.41	62.33	74.69	69.05	74.88
SGFNet [50]	RGB+DSM	91.79	79.18	88.74	62.97	**74.87**	69.47	75.05
PointBoost [14]	RGB + LiDAR	91.05	78.85	85.94	60.85	73.32	59.87	71.77
IKD-Net [15]	RGB + LiDAR	91.45	80.22	87.97	61.77	74.07	70.15	74.84
AMIANet (ours)	RGB + LiDAR	**92.07**	**80.49**	**89.77**	**63.48**	74.15	**70.55**	**75.69**

categories. It achieved the highest IoU in these categories, indicating its ability to effectively utilize key spatial features and significantly improve classification accuracy. For the "Low veg" category, despite the complexity of feature descriptions and the poor performance of many methods, our method still performed excellently, showcasing its strong capability in handling diverse features. In the recognition of the "Tree" category, SS likely due to the spatial structure being easily confused with "Low veg," our method did not perform as well as SGFNet. For the "Car" category, AMIANet achieved a much higher IoU than other methods, demonstrating its ability to accurately identify vehicles in complex backgrounds.

Overall, AMIANet outperformed other algorithms in all detailed metrics, showing significant advantages. Compared to traditional baseline methods, AMIANet achieved higher scores in OA and mIoU. The excellent performance of AMIANet is mainly attributed to its asymmetric network structure and the collaborative multimodal interaction module. The asymmetric structure facilitates balanced information exchange, while the interaction module enables the effective fusion of multimodal features. This design significantly enhances classification accuracy and robustness, allowing AMIANet to excel in processing complex multimodal data.

Fig. 9 shows the segmentation results of AMIANet and several SOTA multimodal segmentation methods in six scenes from the ISPRS Vaihingen dataset. In all scenarios, the multimodal approaches, including AMIANet, demonstrated superior accuracy in extracting feature contours, particularly for buildings with shadows. Notably, in the fourth scenario, most methods misclassified portions of low vegetation as buildings; however, only our method correctly identified these regions. Furthermore, AMIANet excelled in accurately delineating building contours without interference from other categories.

TABLE IV
ABLATION STUDY FOR ASYMMETRICAL STRUCTURE

Variants	N3C-California		WHU-RRDSD	
	mAcc	mIoU	mAcc	mIoU
AMIANet-	81.73	70.02	67.44	60.23
PointNet++	91.20	77.32	**74.95**	67.20
AMIANet(ours)	**92.84**	**78.13**	74.84	**67.51**

F. Ablation Studies

1) Asymmetric Structure Ablation Experiment: The asymmetric network structure achieves a balance in multimodal feature fusion. To demonstrate the effectiveness of the feature preprocessing component in the network, ablation experiments for the asymmetric structure are conducted. Variants include the removal of the feature preprocessing module (AMIANet-) and extraction based on PointNet++ (PointNet++). The experimental results, as shown in Table IV, clearly indicate substantial accuracy improvements on both datasets with the inclusion of the feature preprocessing part. Specifically, the preprocessing of features refines LiDAR characteristics, facilitating multimodal feature interaction. In addition, compared to PointNet++, using RandLA-Net as the extraction backbone yielded better results.

2) MCP Ablation Experiment: Due to the inevitable information mismatch in the projection process, appropriate processing modules can effectively address the issue of hole pixels that arise during projection. To investigate the efficacy

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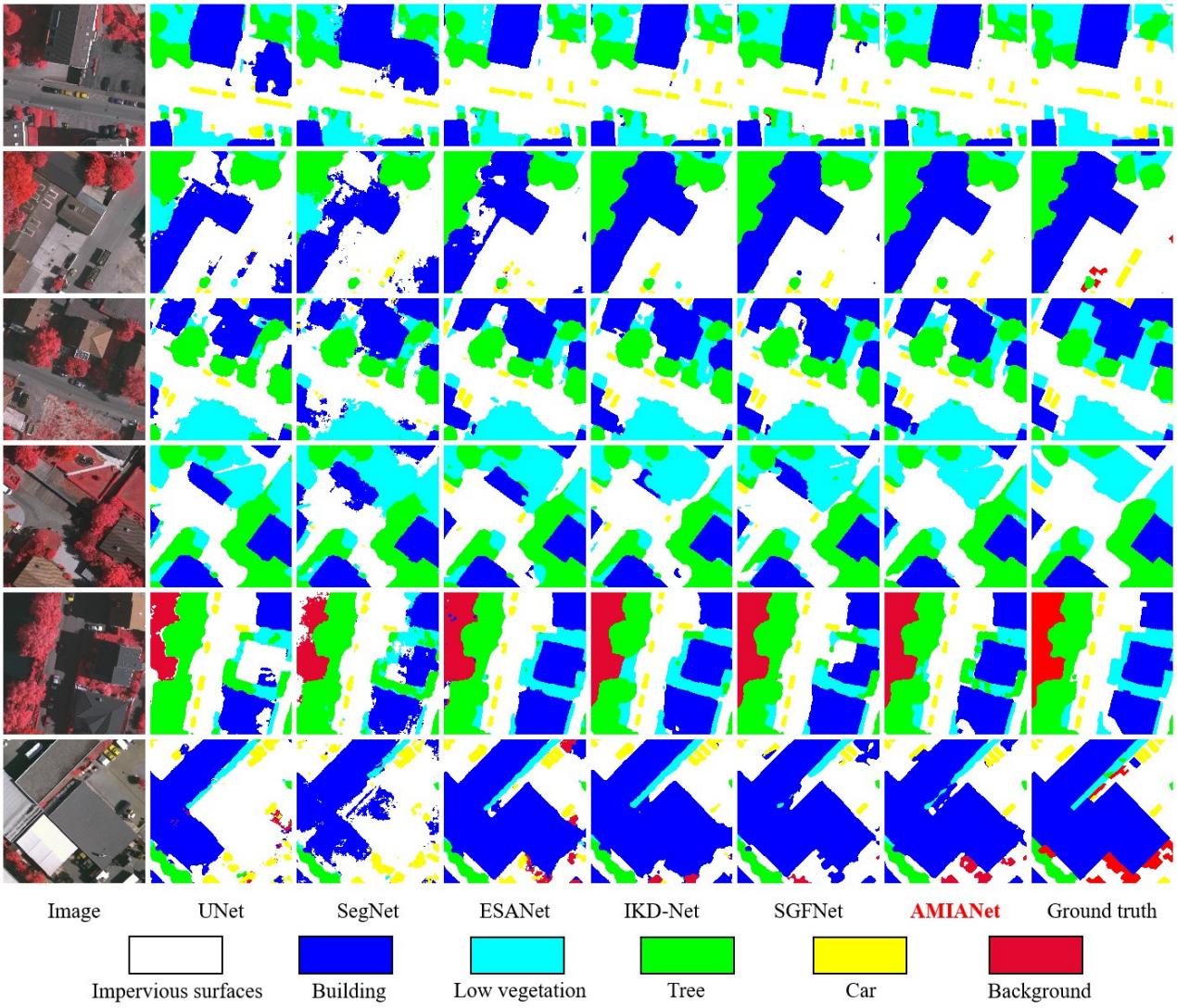

Fig. 9. Qualitative comparison of the baseline, the SOTA multimodal segmentation networks, and AMIANet (ours) on the ISPRS VAIHINGEN DATASET.

of our projection module, ablation experiments are conducted on its components. Different variants included the removal of all components (AMIANet-), retention of only the max pooling (MP), and retention of only the multiscale convolution (MC). The experimental results, as shown in Table V, indicate that MP greatly optimizes the performance during the feature mapping process and resolves the issue of missing pixels. Meanwhile, MC, using different receptive fields, ensures the rationality of filling in missing information. The combination of both components achieved the highest accuracy in different datasets, confirming the effectiveness of the module.

3) SMI Ablation Experiment: An efficient multimodal feature fusion mechanism ensures the full utilization of information from different structures. The SMI module leverages the complementary characteristics of different modalities to facilitate a bidirectional effective flow of information. To explore the role played by different components, a series of variant methods are defined, including the removal of all components to retain only a basic initial fusion (AMIANet-), retention of PDIF without the attention mechanism (PDIF wo Att), retention of PDIF (PDIF), retention of SMI without the attention mechanism (SMI wo Att), and reversal of the order of two modules in SMI (Reversed). As shown in Table VI, the bidirectional feature interaction module performs best in all accuracy metrics. The information from the LiDAR branch provides rich spatial features for imagery, and conversely, the texture information from imagery helps overcome difficulties encountered in LiDAR classification. Moreover, compared to the reverse fusion structure, a logical sequence also ensures the best results.

TABLE V
ABLATION STUDY FOR MCP MODULE

Variants	N3C-California		WHU-RRDSD	
	mAcc	mIoU	mAcc	mIoU
AMIANet-	87.31	76.39	68.09	64.28
MC	90.03	77.18	71.40	65.16
MP	91.29	78.00	73.98	67.20
AMIANet(ours)	**92.84**	**78.13**	**74.84**	**67.51**

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TABLE VI
ABLATION STUDY FOR SMI MODULE

Variants	N3C-California		WHU-RRDSD	
	mAcc	mIoU	mAcc	mIoU
AMIANet-	84.16	76.08	71.14	64.27
PDIF wo Att	90.44	76.31	72.26	65.22
PDIF	91.77	76.65	72.94	65.57
SMI wo Att	91.39	77.79	71.68	66.65
Reversed	91.37	77.92	**75.26**	67.22
SMI (ours)	**92.84**	**78.13**	74.84	**67.51**

TABLE VII
TEST OF SIGNIFICANT DIFFERENCE BETWEEN AMIANET AND SOTA METHODS

Method	N3C-California		WHU-RRDSD	
	mAcc	mIoU	mAcc	mIoU
PACSCNet [46]	0.0006	0.0039	0.0035	0.0038
SGFNet [50]	0.0031	0.0005	0.0042	0.012
PointBoost [14]	0.005	0.0001	0.002	0.0029
IKD-Net [15]	0.0013	0.0021	0.0089	0.004

4) Significance Difference Test: In evaluating segmentation accuracy, the dataset setup described in Section IV-A was utilized. The samples are not independent and may not follow a normal distribution. Therefore, the Wilcoxon signed-rank test was employed to calculate the p-value to verify the superiority of our AMIANet [61]. As the null hypothesis, there is no significant difference in evaluation metrics between AMIANet and the other four SOTA methods (PACSCNet, SGFNet, PointBoost, and IKD-Net) on the N3C-California and WHU-RRDSD datasets. The pairwise Wilcoxon signed-rank tests are performed between AMIANet and each of these four methods, using mAcc and mIoU as the evaluation metrics. According to the alternative hypothesis, significant differences exist between the methods. The experimental results are shown in Table VII.

From Table VII, it can be seen that there is a statistically significant difference between our AMIANet and the other four SOTA methods in terms of mAcc and mIoU. This is consistent with the previous results, confirming that our method achieves the best segmentation performance on both datasets.

5) Multiple Cross Validation: To further demonstrate the robustness of our method, a fivefold cross-validation experiment is conducted on the WHU-RRDSD dataset. The dataset is divided into five parts along the railway line and treated the five parts as separate subsets and performed five runs. Each run used four nonoverlapping subsets for training and the remaining subset for validation. The experimental results are shown in Table VIII, where the results for each metric are accompanied by their corresponding standard deviations.

As shown in Table VIII, our AMIANet achieved the best average performance across all metrics, indicating superior segmentation performance. However, the standard deviations of our method were not always the lowest, which may be due to significant differences in vegetation distribution and architectural styles in different regions along the railway line. Nevertheless, our method demonstrated the best overall performance, exhibiting notable robustness.

TABLE VIII
EXPERIMENTAL RESULTS FOR FIVEFOLD CROSS VALIDATION

Method	OA	mAcc	mIoU
SGFNet [50]	85.52±1.46	72.39±1.96	64.23±1.48
PointBoost [14]	85.34±1.05	71.11±0.85	62.34±0.88
IKD-Net [15]	80.97±0.63	70.45±1.71	60.72±1.04
AMIANet (ours)	**87.13±0.82**	**74.19±1.0**	**65.83±0.77**

V. DISCUSSION

The experimental results on various datasets demonstrate that our method outperforms other multimodal segmentation approaches in terms of segmentation effectiveness. The LiDAR data in the N3C-California dataset possess rich and accurate structural information, with most of the features to be classified having uniform spatial structures and distinct characteristics. Our AMIANet shows superior performance in this scenario. The special projection module ensures the completeness of information during cross-modal conversion, enhancing the utilization of LiDAR features. In addition, the unique network structure and the fusion module optimize the refinement of features from different modalities, as reflected in the higher accuracy for the "Tree" and "Building" categories. Traditional multimodal segmentation methods often focus only on elevation information, neglecting important wall structure and canopy features. Our method integrates spatial structure and texture information, allowing for more precise delineation of corresponding contours.

The WHU-RRDSD dataset features a more diverse range of land objects, with nonuniform spatial structures within the same category, and the "Road" category, being less prominent and having less distinct features, presents significant challenges for multimodal segmentation tasks. SOTA multimodal segmentation methods do not show significant advantages over the baseline method UNet in classifying the "Road" category. Our method, leveraging complementary multimodal information, provides a more accurate basis for discrimination. The preprocessing and interaction of information also ensure decent segmentation precision even when the quality of the LiDAR data is suboptimal.

VI. CONCLUSION

This article introduces a novel end-to-end AMIANet, which preprocesses the LiDAR branch to balance the features of different modalities. Based on the MCP module, the integrity of data during fusion is ensured. A novel SMI module is proposed to achieve bidirectional feature enhancement, utilizing complementary information for a more comprehensive feature representation. Ablation experiments on different datasets were conducted to prove the effectiveness of the aforementioned

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modules, and our method was compared with competitive SOTA methods, demonstrating the superiority of AMIANet. In the future, we aim to optimize the network's structure to achieve high precision and high performance, gaining broader application in tasks such as land cover segmentation. At the same time, an increasing number of UAV platforms are integrating radar and optical sensors, achieving synchronous acquisition of multimodal data through high-precision time synchronization and control systems. Based on this, our method will play an important role in addressing issues such as vegetation cover in crop extraction tasks as well as in fields such as environmental monitoring and disaster assessment.

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